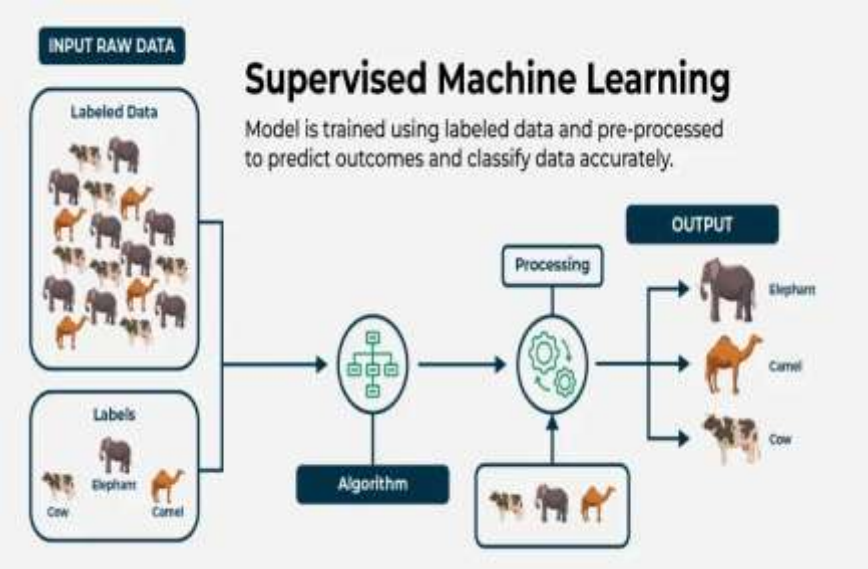
Day- 9 ML Introduction

Machine learning is a branch of Artificial Intelligence that focuses on developing models and algorithms that let computers learn from data without being explicitly programmed for every task. In simple words, ML teaches systems to think and understand like humans by learning from the data.

Supervised Machine Learning:

Supervised learning is a type of machine learning where a model learns from labelled data, meaning each input has a correct output. The model compares its predictions with actual results and improves over time to increase accuracy.



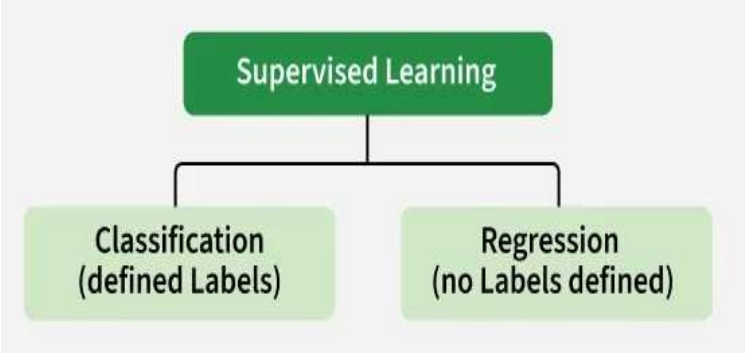
Its main features are:

- Each input has a known output
- Adjusts itself to reduce prediction errors
- Make accurate predictions on new data
- For example it recognizing handwritten digits from trained data

Types of Supervised Learning

Now, Supervised learning can be applied to two main types of problems:

- **Classification:** Where the output is a categorical variable (e.g., spam vs. non-spam emails, yes vs. no).
- **Regression:** Where the output is a continuous variable (e.g., predicting house prices, stock prices).



Working of Supervised Machine Learning:

The working of supervised machine learning follows these key steps:

1. Collect Labeled Data

- Gather a dataset where each input has a known correct output (label).
- **Example**: Images of handwritten digits with their actual numbers as labels.

2. Split the Dataset

- Divide the data into training data (about 80%) and testing data (about 20%).
- The model will learn from the training data and be evaluated on the testing data.

3. Train the Model

- Feed the training data (inputs and their labels) to a suitable supervised learning algorithm (like Decision Trees, SVM or Linear Regression).
- The model tries to find patterns that map inputs to correct outputs.

4. Validate and Test the Model

- Evaluate the model using testing data it has never seen before.
- The model predicts outputs and these predictions are compared with the actual labels to calculate accuracy or error.

5. Deploy and Predict on New Data

- Once the model performs well, it can be used to predict outputs for completely new, unseen data.

Supervised Machine Learning Algorithms

Supervised learning includes different types of algorithms used to predict outputs based on labeled data. Each algorithm is designed for specific tasks like prediction or classification.

- **Linear Regression:** Used to predict continuous values (e.g., price, temperature). It is simple and widely used.

- **Logistic Regression**: Logistic regression is a type of supervised learning classification algorithm that is used to predict a binary output variable.
- **Decision Trees**: Uses a tree-like structure where each node represents a decision and each leaf represents an outcome.
- **Random Forests**: Combines multiple decision trees to improve accuracy and reduce overfitting.
- **Support Vector Machine(SVM):** Separates data into classes using a boundary (hyperplane). Support vectors help define this boundary.
- **K-Nearest Neighbors:** Predicts based on the closest data points. Results depend on the value of k and distance measure.
- **Gradient Boosting:** Builds models step-by-step by correcting errors of previous models, creating a strong model.
- **Naive Bayes Algorithm:** Based on probability and Bayes' Theorem, assuming features are independent of each other.

Examples

- **Fraud Detection in Banking**: Uses labeled transaction data to identify and predict fraudulent activities.
- **Parkinson Disease Prediction:** Analyzes medical data such as voice recordings, motor symptoms, and clinical measurements to identify patterns and predict whether a patient has disease or not.
- **Customer Churn Prediction:** Uses historical customer data to predict whether a customer will leave a service.
- **Cancer cell classification:** Implements supervised learning for cancer cells based on their features and identifying them if they are 'malignant' or 'benign.
- **Stock Price Prediction**: Uses past data to predict stock trends and support investment decisions.

Advantages:

- Easy to understand and implement as it learns from labeled data.
- Provides high accuracy when sufficient labeled data is available.
- Works for both classification (spam detection, disease prediction) and regression (price forecasting).
- Can generalize well to unseen data with proper training and diverse datasets.
- Widely used in applications like speech recognition, medical diagnosis, sentiment analysis and fraud detection.

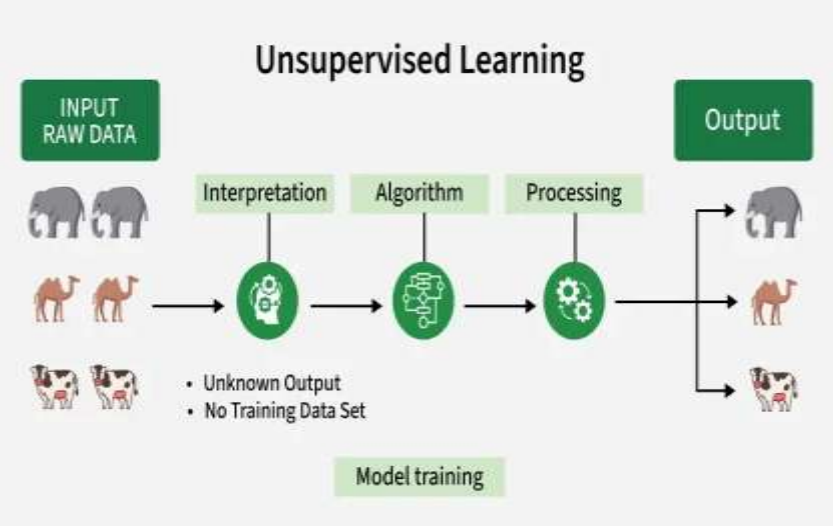
Disadvantages:

- Requires large amounts of labeled data, which is expensive and time-consuming to prepare.
- Can be biased if the training data is unbalanced, leading to unfair or inaccurate predictions.
- May overfit the training data instead of learning general patterns, especially with small datasets.
- Performance can drop when applied to data that is very different from the training data.
- Not easily scalable for problems with a very large number of labels, such as in natural language tasks.

Unsupervised Machine Learning

Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.

- It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.
- Helps identify hidden patterns in data
- Useful for grouping, compression and anomaly detection



Working of Unsupervised Learning

The working of unsupervised machine learning can be explained in these steps:

1. Collect Unlabeled Data

- Gather a dataset without predefined labels or categories.
- **Example**: Images of various animals without any tags.

2. Select an Algorithm

- Choose a suitable unsupervised algorithm such as clustering like K-Means, association rule learning like Apriori or dimensionality reduction like PCA based on the goal.

3. Train the Model on Raw Data

- Feed the entire unlabeled dataset to the algorithm.
- The algorithm looks for similarities, relationships or hidden structures within the data.

4. Group or Transform Data

- The algorithm organizes data into groups (clusters), rules or lower-dimensional forms without human input.
- Example: It may group similar animals together or extract key patterns from large datasets.

5. Interpret and Use Results

- Analyze the discovered groups, rules or features to gain insights or use them for further tasks like visualization, anomaly detection or as input for other models.

Unsupervised Learning Algorithms

There are mainly 3 types of Unsupervised Algorithms that are used:

1. Clustering Algorithms

Clustering is an unsupervised machine learning technique that groups unlabeled data into clusters based on similarity. Its goal is to discover patterns or relationships within the data without any prior knowledge of categories or labels.

- Groups data points that share similar features or characteristics.
- Helps find natural groupings in raw, unclassified data.
- Commonly used for customer segmentation, anomaly detection and data organization.
- Works purely from the input data without any output labels.
- Enables understanding of data structure for further analysis or decision-making.

Some common clustering algorithms:

- **K-means Clustering:** Groups data into K clusters based on how close the points are to each other.
- **Hierarchical Clustering:** Creates clusters by building a tree step-by-step, either merging or splitting groups.
- **Density-Based Clustering (DBSCAN):** Finds clusters in dense areas and treats scattered points as noise.

- **Mean-Shift Clustering:** Discovers clusters by moving points toward the most crowded areas.
- **Spectral Clustering:** Groups data by analyzing connections between points using graphs.

2. Association Rule Learning

Association rule learning is a rule-based unsupervised learning technique used to discover interesting relationships between variables in large datasets. It identifies patterns in the form of "if-then" rules, showing how the presence of some items in the data implies the presence of others.

- Finds frequent item combinations and the rules connecting them.
- Commonly used in market basket analysis to understand product purchase relationships.
- Helps retailers design promotions and cross-selling strategies.

3. Dimensionality Reduction

Dimensionality reduction is the process of decreasing the number of features or variables in a dataset while retaining as much of the original information as possible. This technique helps simplify complex data making it easier to analyze and visualize. It also improves the efficiency and performance of machine learning algorithms by reducing noise and computational cost.

- It reduces the dataset's feature space from many dimensions to fewer, more meaningful ones.
- Helps focus on the most important traits or patterns in the data.
- Commonly used to improve model speed and reduce overfitting.

Here are some popular Dimensionality Reduction algorithms:

- **Principal Component Analysis (PCA):** Reduces dimensions by transforming data into uncorrelated principal components.
- **Non-negative Matrix Factorization (NMF):** Breaks data into non-negative parts to simplify representation.
- **Locally Linear Embedding (LLE)**: Reduces dimensions while preserving the relationships between nearby points.
- **Isomap**: Captures global data structure by preserving distances along a manifold.

Applications

- **Customer Segmentation**: Algorithms cluster customers based on purchasing behavior or demographics, enabling targeted marketing strategies.
- **Anomaly Detection**: Identifies unusual patterns in data, aiding fraud detection, cybersecurity and equipment failure prevention.

- **Recommendation Systems**: Suggests products, movies or music by analyzing user behavior and preferences.
- **Image and Text Clustering**: Groups similar images or documents for tasks like organization and content recommendation.
- **Social Network Analysis**: Detects communities or trends in user interactions on social media platforms.

Advantages

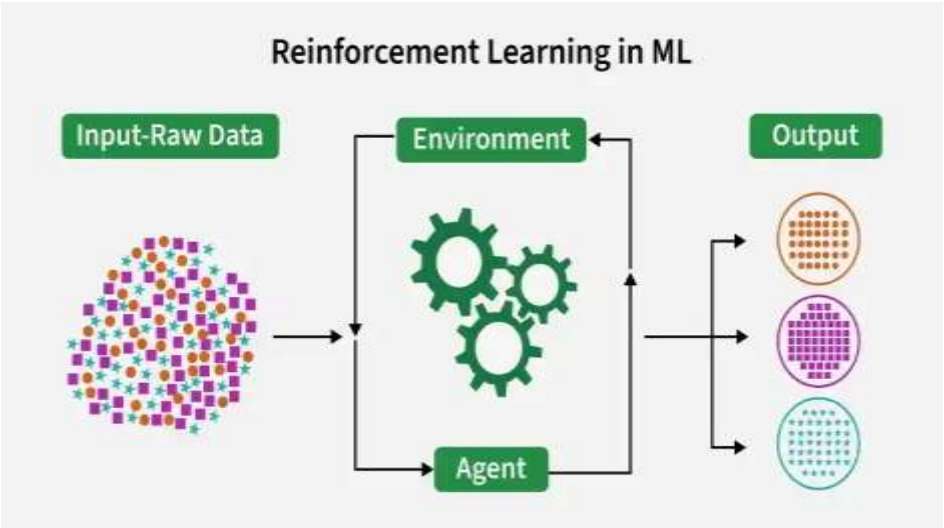
- Works with raw, unlabeled data, saving time and effort required for data annotation.
- Finds hidden patterns and natural groupings in data that may not be easily identified by humans.
- Handles large and complex datasets efficiently, including high-dimensional data.
- Helps detect anomalies and unusual data points without needing prior examples.

Challenges

- Noisy data and outliers can distort patterns and reduce the effectiveness of the model.
- Models may capture noise instead of meaningful patterns, leading to overfitting.
- Lack of labeled data makes it difficult to guide the algorithm toward specific outcomes.
- Results such as clusters may be difficult to interpret or may not clearly match real-world categories.

Reinforcement Learning

Reinforcement Learning (RL) is a branch of machine learning that focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards. RL allows machines to learn by interacting with an environment and receiving feedback based on their actions. This feedback comes in the form of rewards or penalties.

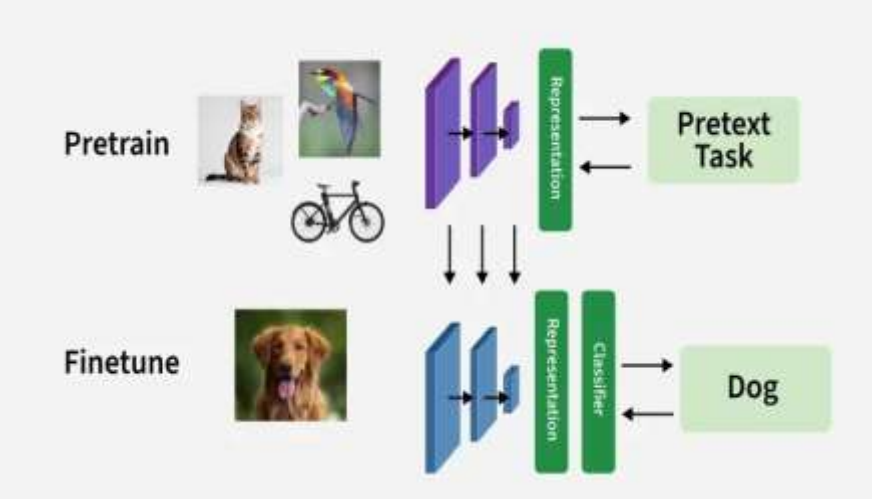


Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal. The agent performs actions and receives feedback to optimize its decision-making over time.

- **Agent**: The decision-maker that performs actions.
- **Environment**: The world or system in which the agent operates.
- **State**: The situation or condition the agent is currently in.
- **Action**: The possible moves or decisions the agent can make.
- **Reward**: The feedback or result from the environment based on the agent's action.

Self-Supervised Learning (SSL)

Self-Supervised Learning (SSL) is a type of machine learning where a model is trained using data that does not have any labels or answers provided. Instead of needing people to label the data, the model finds patterns and creates its own labels from the data automatically.



Semi-Supervised Learning in ML

Semi-supervised learning is a distinct machine learning approach that uses a small amount of labeled data along with a large amount of unlabeled data to improve model performance. The goal is to learn a function that accurately predicts outputs based on inputs, similar to supervised learning, but with much less labelled data.

